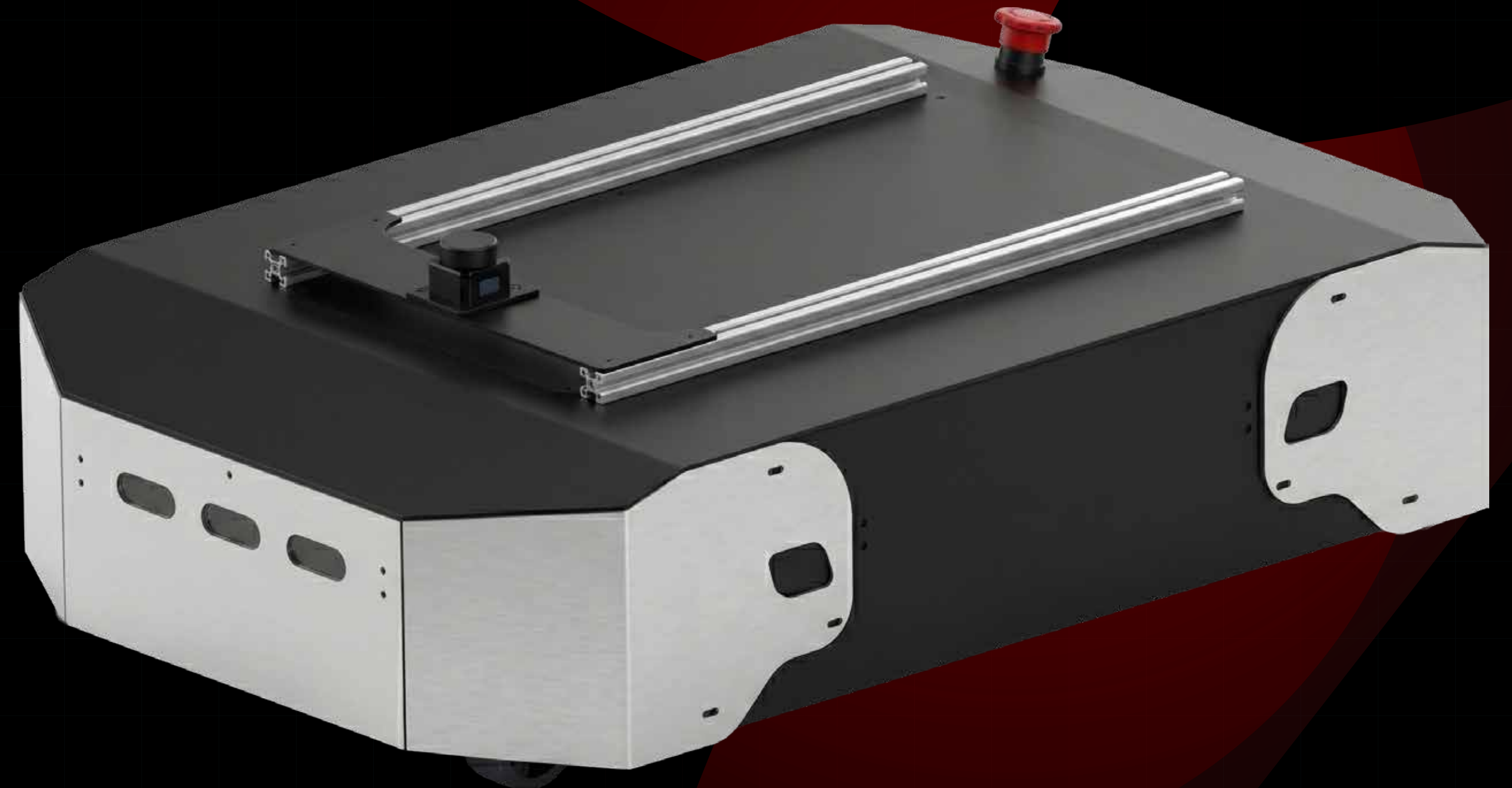


Cruiser 2.0

High-Payload Intelligent
UGV Platform

Advanced Research,
and Industrial Logistics Base



Overview & Objectives

The Cruiser 2.0 is a high-performance, two-wheel differential Unmanned Ground Vehicle (UGV) designed for university research and industrial prototyping. It bridges the gap between a classroom teaching tool and a functional logistics platform, offering a robust 100kg payload capacity.

Heavy-Duty Research:

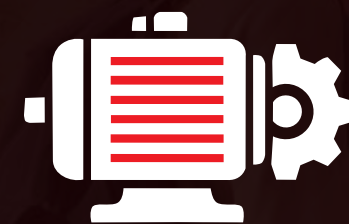
Capable of carrying full-sized robotic arms (cobots), heavy sensor arrays, or logistics payloads up to 100kg.

Technical Specifications

Performance & Mechanical



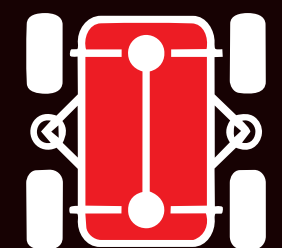
Rated Payload
100 kg
Full Operational Load



Actuation
High-torque DC Hub Motors (High-precision BLDC)



Speed & Mobility
Optimized for high-load stability and lab safety



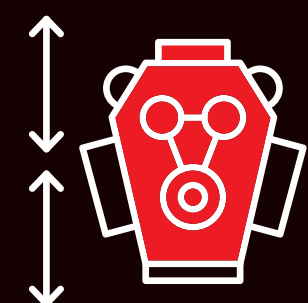
Drive Type
Two-wheel differential drive with 4 passive caster wheels



Safety
Rear E-Stop Switch, Front Illumination LEDs



Expansion
Top-mounted "Bayonet Open Compartment" for payloads



Robot dimensions
85x60x25cm



max speed
1m/s



weight
50kg

Technical Specifications

Advanced Sensor Suite



LiDAR:

360° Laser Radar (LD06), 12m Range, 4500Hz



IMU:

9-Axis (Absolute orientation /Sensor fusion)



Odometry:

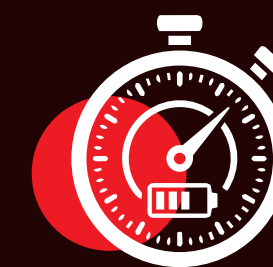
High-resolution Hall-effect Encoders (built-in motors)

Power System



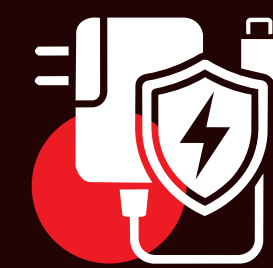
Battery Type:

36V Lithium-ion Battery Set (High Energy Density)



Endurance:

Optimized for long-duration laboratory experiments



Charging:

42V 3A Smart Charger (Approx. 6–8 hours full charge)

Software



Framework:

Ubuntu 22.04 LTS + ROS 2 Humble Hawksbill

Educational Applications

The Cruiser 2.0 is designed to support a multi-year robotics curriculum,

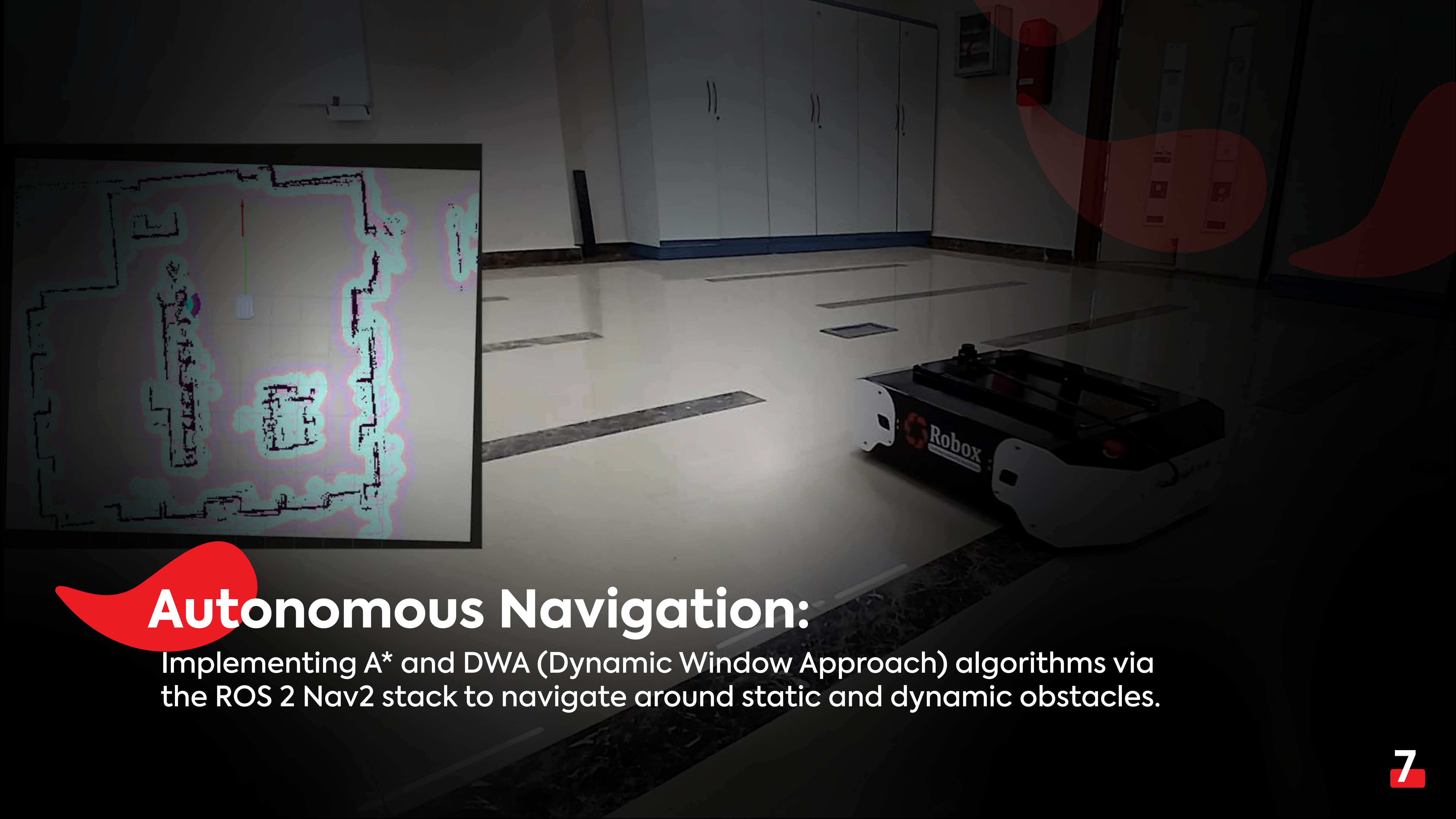


Kinematic Modeling:

Exploring Forward and Inverse Kinematics (Wheel velocities to Robot motion).

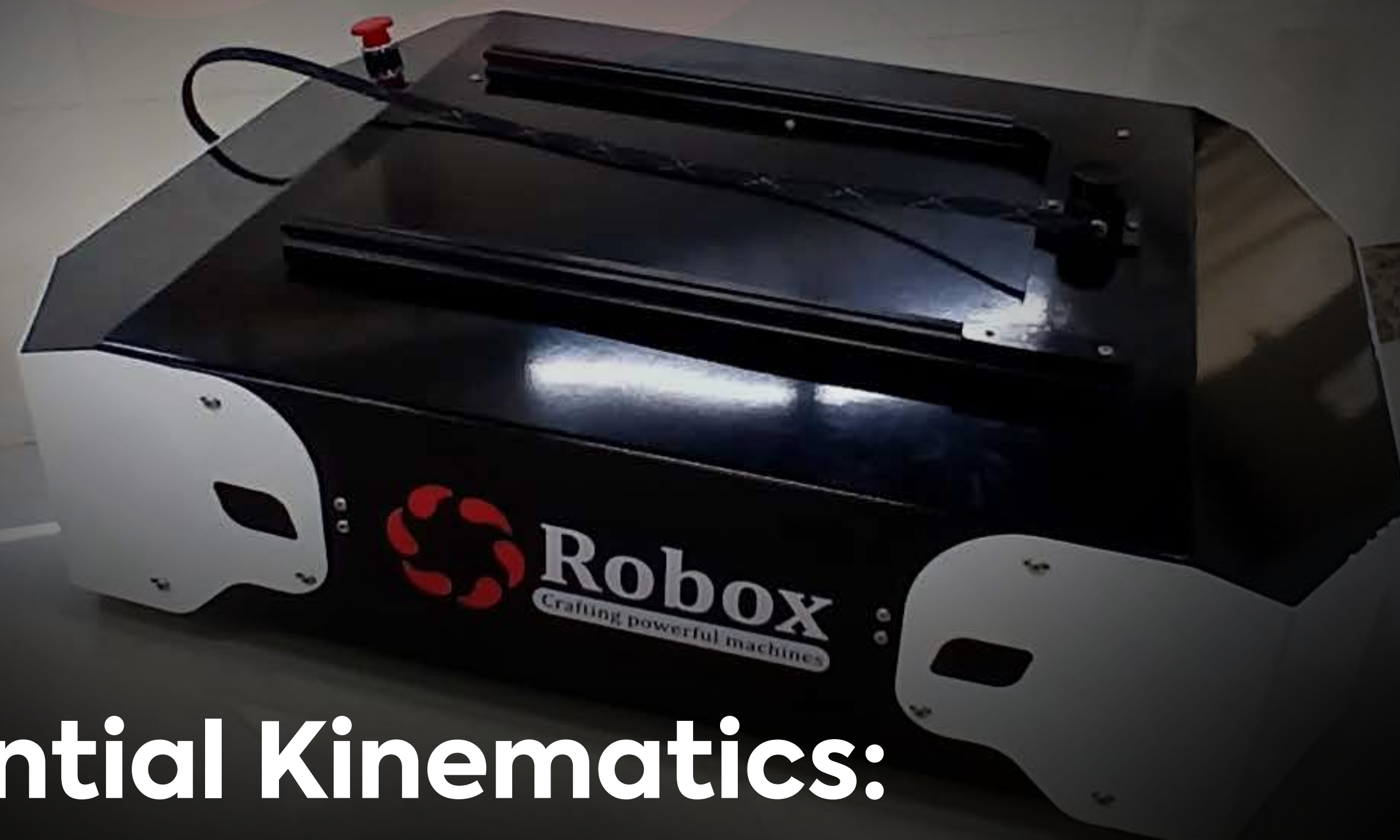
Odometry & Sensor Fusion:

Using Extended Kalman Filters (EKF) to fuse noisy wheel encoder data with IMU data for "Dead Reckoning" localization.



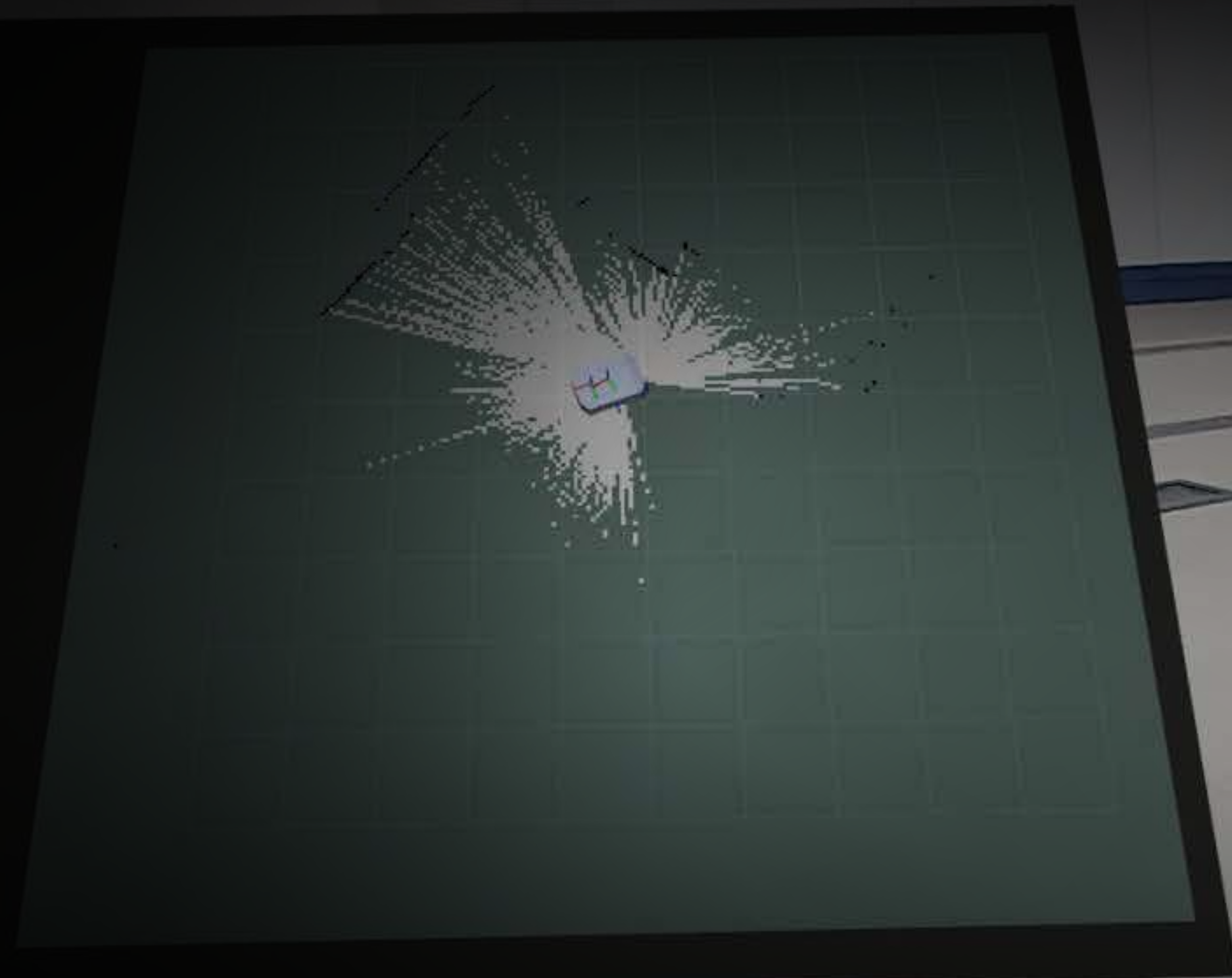
Autonomous Navigation:

Implementing A* and DWA (Dynamic Window Approach) algorithms via the ROS 2 Nav2 stack to navigate around static and dynamic obstacles.



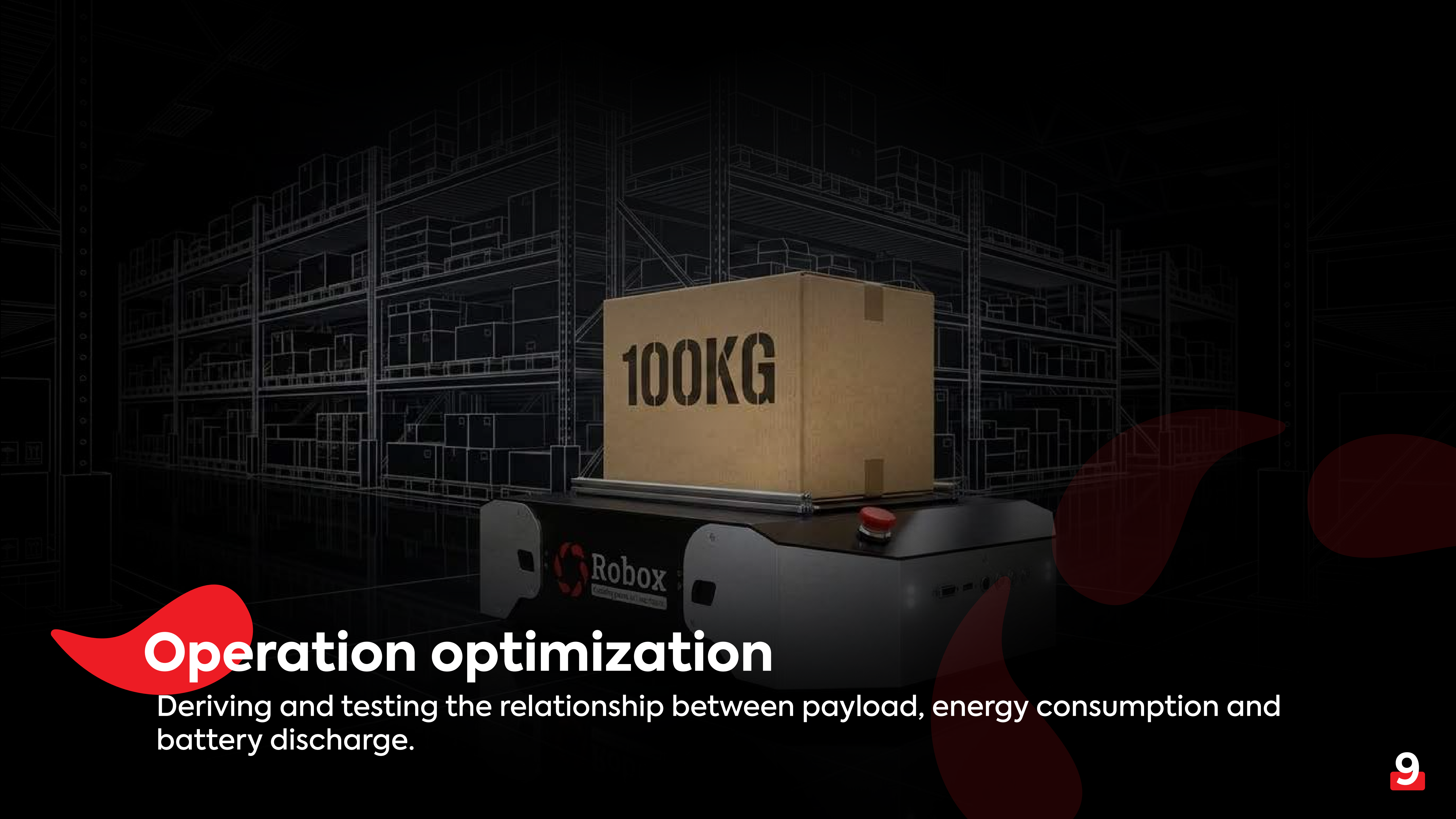
Differential Kinematics:

Deriving and testing the relationship between wheel RPM and linear/angular velocity.



Mapping & SLAM:

Utilizing SLAM Toolbox to create high-definition 2D maps of indoor environments.



Operation optimization

Deriving and testing the relationship between payload, energy consumption and battery discharge.

Scope of Delivery

The ArmBot kit is a comprehensive, “plug-and-play” educational solution. Each unit is shipped fully assembled and includes the following components:

Cruiser 2.0 Base

High-strength UGV chassis with integrated Hub Motors and 100kg load rating.

Computing Unit

Computer unit pre-configured with Ubuntu 22.04 and ROS 2 Humble.

Perception Kit

360° LiDAR and integrated 9-Axis IMU for SLAM and localization.

Control Kit

PS4 Wireless Joystick for manual teleoperation and safety override.

Development Kit

Full access to C++ Chassis Drivers, Python APIs, and ROS 2 workspace.

Educational Pack

Comprehensive User Manual, sample navigation code, and lab tutorials.





Crafting powerful machines
Contact us NOW!

 info@robox-industries.com

 +201210008719

 Robox-industries.com

 facebook.com/robox.industries

 linkedin.com/company/robox-industries

QR

